Awkward Arrays: Accelerating scientific data analysis on irregularly shaped data

Framework, OAC-2103945   PI: J. Pivarski (P. Elmer)

What Awkward Array does

Example of an "awkward" array

```
array = ak.Array([
    [{"x": 1.1, "y": [1]}, {"x": 2.2, "y": [1, 2]}, {"x": 3.3, "y": [1, 2, 3]}],
    [],
    [{"x": 4.4, "y": [1, 2, 3, 4]}, {"x": 5.5, "y": [1, 2, 3, 4, 5]}]
])
```

Record structures with differently typed fields

Array of variable-length lists ("ragged" or "jagged" arrays)

Nested variable-length lists

Missing data

Heterogenous data (union/variant types)

NumPy-like expression

```
output = np.square(array["y", ..., 1:])
```

Result

```
output.to_list()
    [
        [[], [4], [4, 9]],
        [],
        [[4, 9, 16], [4, 9, 16, 25]]
    ]
```

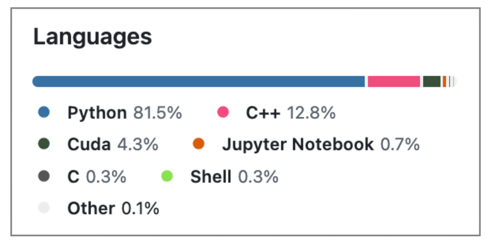
Equivalent Python

```
output = []
for sublist in python_objects:
    tmp1 = []
    for record in sublist:
        tmp2 = []
        for number in record["y"][1:]:
            tmp2.append(np.square(number))
        tmp1.append(tmp2)
    output.append(tmp1)
```

(for a similar calculation 10 million times larger, single threaded)

1.5 seconds
2.1 GB of memory

140 seconds
22 GB of memory



Awkward Releases Highlights (Aug 2024 - now)

- Named axis support for ak.Array - making Awkward Array's multidimensional irregular data manipulation readable and robust
- Virtual Arrays support - "read what you need"
- Enhanced error reporting and performance improvements
- Improved and more complete GPU support - CUDA backend
- Major progress on segmented reducers: argmin, argmax, count_nonzero
- Expanded Tensor framework support (TensorFlow, PyTorch, JAX, cuDF)
- Continuous compatibility maintenance (Python 3.13, Numpy 2.3, etc.)

*An array library for **nested**, **variable-sized data**, including arbitrary-length lists, records, mixed types, and missing data, using **NumPy-like idioms**.*

Awkward Array

*Arrays are **dynamically typed**, but operations on them are **compiled and fast**.*
Coincides with NumPy when arrays are regular; generalizes when they're not.

Awkward Array Library Integrations

Awkward Array is a part of larger scientific Python ecosystem. It must work well with other libraries.

Lossless, bidirectional, zero-copy conversions between Awkward Array, Apache Arrow, Feather, and Parquet.

JIT-compiled as a C++ iterator in ROOT's RDataFrame workflow.

JIT-compile Awkward Arrays in C++ with cppyy. Header-only library to build Awkward Arrays in C++ and transfer them to Python.

Manipulating ragged arrays as though they were **NumPy** or **CuPy** arrays, following the Array API specification.

Auto-differentiate through expressions involving Awkward Arrays, using JAX

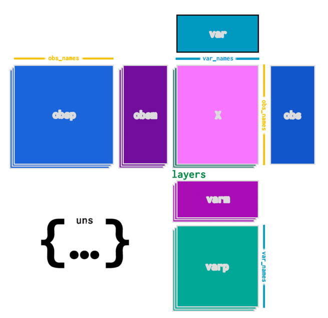
Efficiently iterate over or build Awkward Arrays in code JIT-compiled with Numba, on CPUs and GPUs

Bidirectional conversion between Awkward Array and TensorFlow RaggedTensor.

Store Awkward Arrays in the Tiled database and slice them in the cloud.

Awkward Arrays on GPUs, backed by CuPy.

dask-awkward is a high-level collection type for Dask, alongside Array and DataFrame.



Integrated into AnnData for single-cell genomics.

Read any Kaitai Struct format into Awkward Arrays.

Akimbo: work with Awkward Arrays as DataFrame columns

Exchange data with the Julia language through a reimplementation of the Awkward Array memory layouts.

Adoption

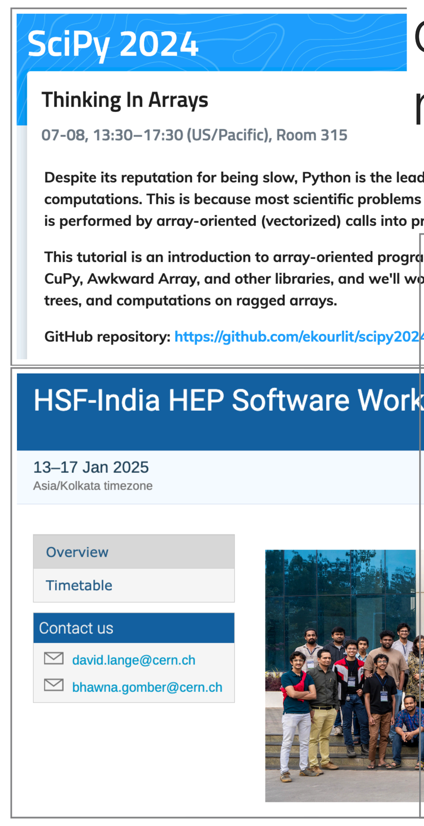
Awkward Array was developed in the context of particle physics

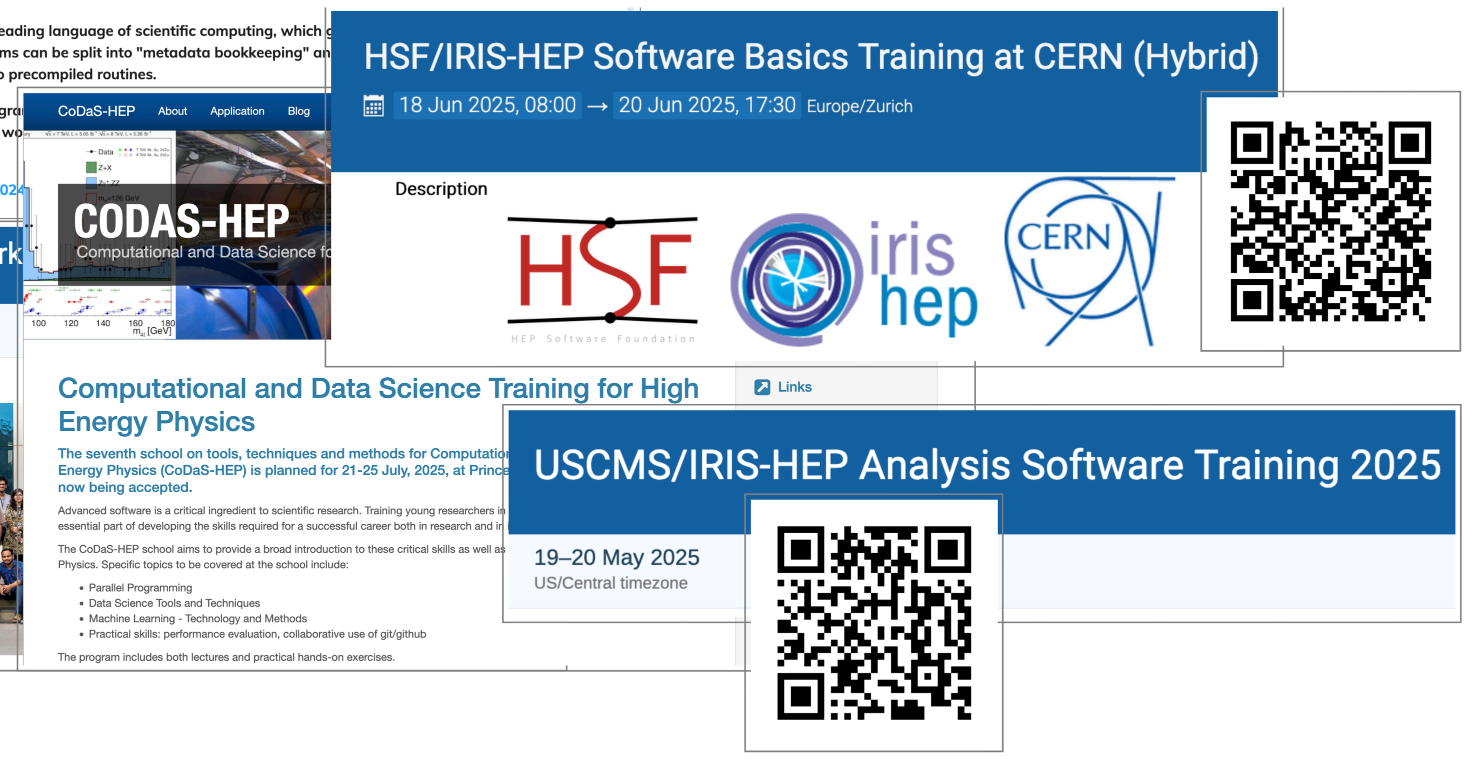
"Non-HEP" use-cases include

- Neutrinos: P-ONE, CIEMAT-Neutrino, Harvard-Neutrino, LEGEND, DUNE, ICARUS, IceCube, nEXO, GraphNeT
- Statistics: errors in annotated data (UKPLab/nessie), Baysean analysis (bat/batty), cmpatino/optimal-observables multivariate shapelets (bianchimario/MARS), text embedding (taylorai/mlx_embedding_models)
- Climate science: Lagrangian ocean probes (Cloud-Drift/clouddrift), cloud microphysics (yoctoyotta1024/CLEO)
- Biology: single-cell genetics (scverse), genotyping (kage-genotyper/kage), astrocytes (janreising/astroCAST)
- Astronomy: active galactic nuclei (Zstone19/TempMap), dark matter direct detection (XENONnT/fuse)
- Defense: identifying missile launches through ionic disturbances in GPS (tylerni7/missile-tid)
- Other:
  - Simulation of fire (silvxlabs/DripTorch),
  - Food orders (EASS-HIT-PART-A-2022-CLASS-II/food-ordering),
  - Analyzing lastFM data (delannoy/lastfm-explorer),
  - Software (WayScience/software-landscape-analysis),
  - Polygon meshes (dewloosh/PolyMesh),
  - CAD (j8sr0230/codelink, AxisVM/pyaxisvm),
  - adarshchbs/pdf2md

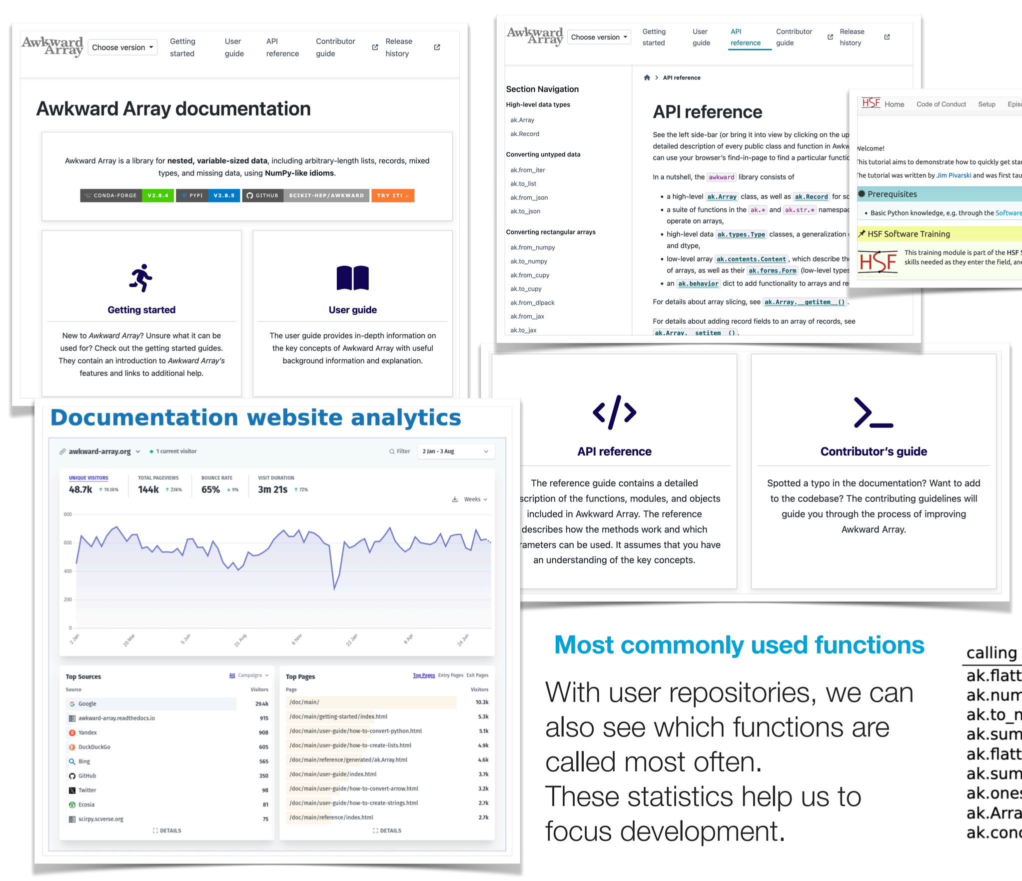
Training, Education and Outreach

The most recent events that include teaching columnar data analysis with Awkward Array in interactive Jupyter notebooks.





Documentation



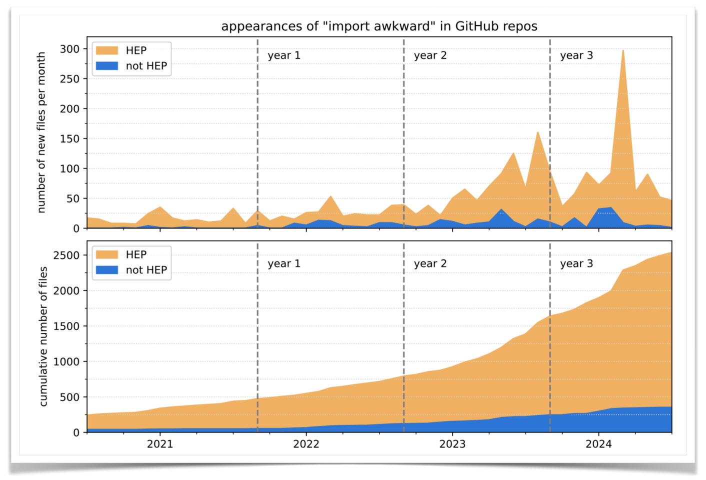
Most commonly used functions

With user repositories, we can also see which functions are called most often. These statistics help us to focus development.

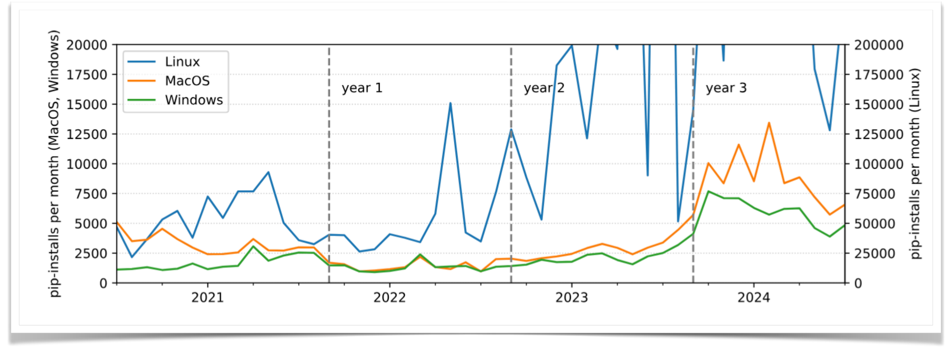
calling pattern	count
ak.flatten(·)	2832
ak.num(·)	2498
ak.to_numpy(·)	2193
ak.sum(·, axis=1)	874
ak.flatten(·, axis=None)	865
ak.sum(·)	564
ak.ones_like(·)	455
ak.Array(·)	406
ak.concatenate(·)	283

function	#unique visitors	#views	avg. time
ak.Array	785	1100	3m33s
ak.concatenate	223	293	4m35s
ak.count	210	265	4m20s
ak.flatten	203	242	4m23s
ak.where	202	262	3m54s
ak.num	184	235	3m07s
ak.to_numpy	181	218	3m25s
ak.mask	178	231	3m52s
ak.zip	163	221	5m02s

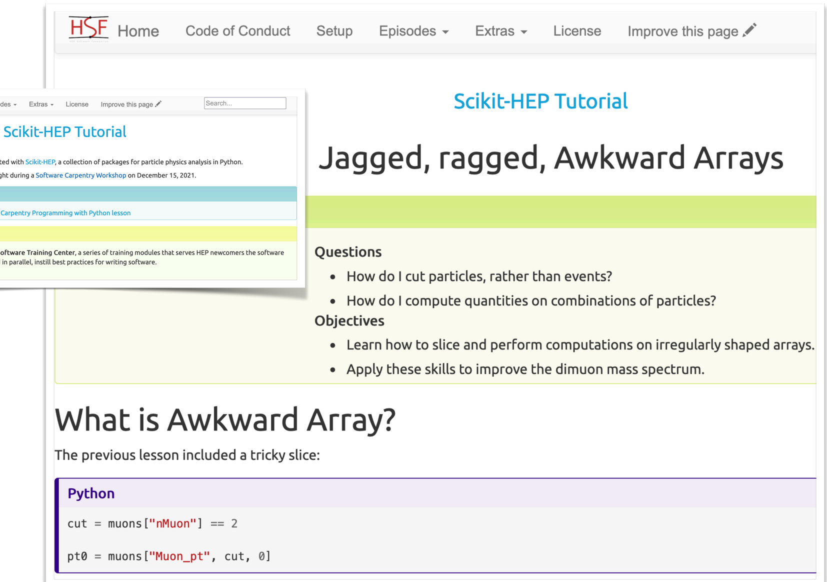
Analysis of user code
We clone all our users' non-fork repositories and search for "import awkward" in Python and Jupyter files.



Download statistics
Frequency of "pip install awkward" is affected by batch jobs and automated testing that installs Awkward Array as a first step, but more so on Linux than MacOS and Windows.

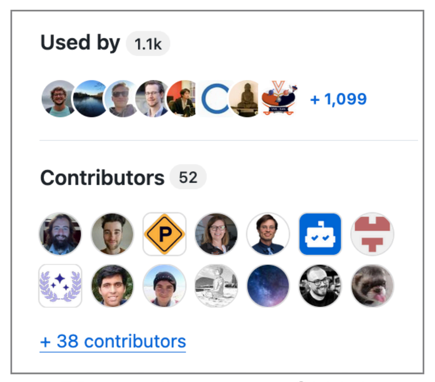


Self-study Tutorials



Team

Regular contributors to Awkward Array during the time of this grant, September 2021 – present.



52 contributors on GitHub